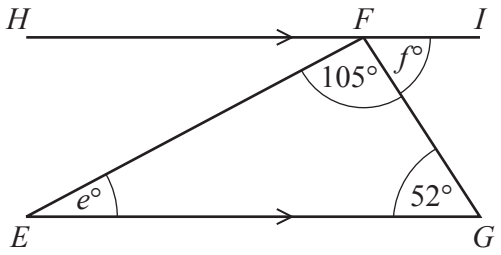
5 **(a)**



NOT TO SCALE

The diagram shows a triangle *EFG* and a straight line *HFI*.
HFI is parallel to *EG*.

(i) Angle *EFG* = 105°.

Write down the mathematical name for this type of angle.

..... [1]

(ii) Work out the value of *e*.
Give a geometrical reason for your answer.

e = ... because .....

..... [2]

(iii) Find the value of *f*.
Give a geometrical reason for your answer.

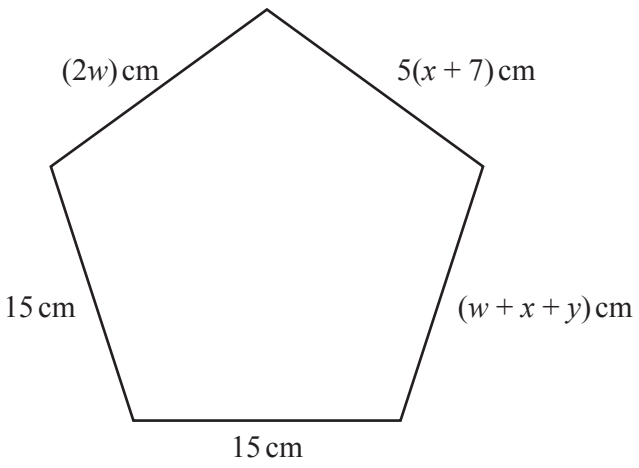
f = ... because .....

..... [2]

(b) Calculate the interior angle of a regular 7-sided polygon.
Give your answer correct to 2 decimal places.

..... [3]

(c) The diagram shows a regular pentagon with sides 15 cm.



NOT TO SCALE

Work out the values of w, x and y.

$w =$ .....

$x =$ .....

$y =$ ..... [5]